

José Ferreira

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EDUCATION

PhD in Physics

University of Aveiro

Supervisors: Miguel Zilhão, Carlos Herdeiro and Raimon Luna

08/2023 - PRESENT

Master in Physics

Faculty of Sciences of the University of Lisbon

Thesis: Constraining $f(Q)$ Cosmology with Standard Sirens [\[arXiv\]](#)

Supervisors: Nelson Nunes and Tiago Barreiro

09/2020 - 09/2022

Bachelor in Physics

Faculty of Sciences of the University of Lisbon

09/2017 - 07/2020

RESEARCH INSTITUTES

Gr@v

Member of the Gravitation group integrated in the Center for Research & Development in Mathematics and Applications (CIDMA) of the University of Aveiro, Portugal.

08/2023 - PRESENT

Institute of Astrophysics and Space Sciences

Member of the cosmology group at the Institute of Astrophysics and Space Sciences, Lisbon node, Portugal.

09/2021 - 07/2023

TEACHING

Assistant

Faculty of Sciences of the University of Lisbon

Courses: Experimental Physics I, Elements of Physics, Mathematical Methods for Physicists, General Physics

02/2022 - 07/2023

PUBLICATIONS

[1] José Ferreira, Tiago Barreiro, José Mimoso, and Nelson J. Nunes. “Testing Λ -free $f(Q)$ cosmology”. *Phys. Rev. D* 108 (6 Sept. 2023), p. 063521. doi: [10.1103/PhysRevD.108.063521](https://doi.org/10.1103/PhysRevD.108.063521).

[2] José Ferreira, Tiago Barreiro, José Mimoso, and Nelson J. Nunes. “Forecasting $F(Q)$ cosmology with Λ CDM background using standard sirens”. *Phys. Rev. D* 105 (12 June 2022), p. 123531. doi: [10.1103/PhysRevD.105.123531](https://doi.org/10.1103/PhysRevD.105.123531).

COMMUNICATIONS

CONTRIBUTED TALKS

[1] “Non-linear Superradiance of Charged Black Holes: Beyond Spherical Symmetry”. 18th-22th of November, 2024. VI Amazonian Symposium on Physics, Belém, Brazil. (Links: [pptx](#), [pdf](#))

[2] “Using Standard Sirens to test non-metric theories of gravity”. 19th-20th of December, 2023. XVI Black Holes Workshop, Porto, Portugal. (Links: [pptx](#), [pdf](#))

[3] “Forecasting $F(Q)$ Cosmology with Λ CDM Background using Standard Sirens”. 4th-6th of May, 2022. Iberian Cosmology Meeting 2022, Barcelona, Spain. (Links: [recording](#), [pdf](#), [pptx](#))

SEMINARS

[1] “Cosmological Tests on Non-Metricity Based Theories of Gravity”. 20th of September, 2023. Gr@v, Aveiro, Portugal. (Links: [recording](#), [pptx](#), [pdf](#))

POSTERS

[1] “Testing Λ -Free $f(Q)$ Cosmology”. 3th-5th of May, 2023. Dark Matter and Stars, Lisbon, Portugal. (Links: [pdf](#))

High Level Languages Julia, Python, Mathematica

Low Level Languages C, Fortran, CUDA

Besides programming skills, I am a regular Linux user, with years of experience using it on personal machines and on servers. I work mostly on a terminal based ecosystem, creating and using terminal utilities to solve personal and academic projects. I am able of creating basic websites using tools such as Quarto or Hugo, and perform basic audio, video and image editing.

AWARDS

Poster award 2023

Dark Matter and Stars 2023

Award for the poster "Testing Λ -Free $f(Q)$ Cosmology" at the 2023 edition of the conference Dark Matter and Stars.

Academic Merit Diploma 2019

Faculty of Sciences of the University of Lisbon

An honorable mention awarded in my second year as an undergraduate student.

Academic Merit Award 2018

Faculty of Sciences of the University of Lisbon

An award for having the best grade in the first year of my undergraduate course.

ACADEMIC ACTIVITIES

Supervisor at the IAstro Summer Internships 2023 07/2023

Institute of Astrophysics and Space Sciences

Supervisor of two undergraduate projects that consisted on constraining modified gravity cosmological models with current and forecast datasets coming from different sources.

Podcast "Physically Speaking" 12/2020 - 12/2022

 [Fisicamente Falando](#)  [Fisicamente Falando](#)  [fisicamentefalando.nfef](#)  [fisicamente_falando_](#)

A student driven podcast created during the pandemic where we interviewed faculty members to speak about their lives, both inside and outside of academia. I am in charge of the audio and video editing and also take part in some of the interviews.

Member of the Student Body for Physicists and Physical Engineers 10/2020 - 10/2021

 [nfef-fcul.com](#)  [NFEFFCUL](#)  [nfeffcul](#)

I was in charge for the development of the digital tools of the student body of physicists and physical engineers. This included the website, the online repository (which is made up of student provided content) and the Discord channel.

OUTREACH

European Researchers' Night 2024 27/09/2024

Coordinator of the participation of Gr@v in the European Researcher's Night in Aveiro, Portugal. Using digital content, we explained the physics behind Gravitational Waves and lensing around compact objects.

"Cosmic Jungle: Searching for Black Holes" 03/05/2024

 [Presentation \(.pptx\)](#)

Presentation to high school students at the event *XPerimenta2024*, in the University of Aveiro, on the properties of black holes and how to find them.

Project "Physics on the loose" 01/2023 - 07/2023

A project, hosted at the Faculty of Sciences of the University of Lisbon, with a room full of interactive experiments to illustrate several fundamental concepts in Physics.

"When a Black Hole Falls Inside... Another Black Hole" 15/04/2023

 [Presentation \(.pptx\)](#)

Presentation of an Ignite style talk in an event organized by the Institute of Astrophysics and Space Sciences that brings space sciences to cities far away from the main urban areas.